

Calibres in GroupLab

The calibre is entered as the bullet's diameter, and nothing else. Generated by grouplab calibres from `src/GroupLab.Core/Marking/Calibre.cs`; do not edit it by hand, because a test fails when this file differs from what the command writes.

1. How to enter a calibre

Type the bullet diameter in inches, such as .308 or 0.308, or in millimetres with mm, such as 7.82 mm. A number marked in or with an inch mark is inches too. A bare number of one or more, such as 7.62, 308 or 22, is refused rather than guessed, because a calibre's name is usually not its bullet's diameter: read as millimetres, 7.62 would be 0.300 in, which no 7.62 bullet is. Names such as 300 Blackout, 6.5 Creedmoor or 30 Cal. are refused for the same reason. Every diameter must lie between 0.1 and 1 in.

A calibre's designation is refused in either unit, because it is a name written as a number: typed as a diameter, .38 would be 0.380 in when a .38 bullet is .357 or .358, and 7.62 mm would be 0.300 in. A value typed with more digits is the same value, so .270 is refused with .27. The designations refused are, in inches, .17, .20, .22, .25, .27, .28, .30, .303, .32, .35, .38, .44 and .45, and in millimetres, 5.45 mm, 5.56 mm, 6 mm, 6.5 mm, 6.8 mm, 7 mm, 7.5 mm, 7.62 mm, 7.65 mm, 8 mm, 9 mm and 10 mm. Those that are also real diameters are not refused: .40, .41 and .50 in, and 9.3 and 12.7 mm. Nothing is read from these lists; they only refuse.

A designation's refusal names the problem, for example: "7.62 mm is a calibre's name, not the bullet's diameter. Enter the bullet's diameter, such as 7.82 mm or 0.308."

A refusal says what to type: "Enter the bullet diameter in inches, such as 0.308, or in millimetres with mm, such as 7.82 mm."

2. The diameters in the pick list

Every distinct bullet diameter in Alan's rifle and pistol lists, smallest first. Any other diameter in range can be typed.

Inches	Millimetres
.172	4.37
.204	5.18
.2215	5.63
.224	5.69
.243	6.17
.257	6.53
.264	6.71
.277	7.04
.284	7.21
.308	7.82
.309	7.85
.310	7.87
.3105	7.89

.312	7.92
.321	8.15
.323	8.20
.338	8.59
.355	9.02
.356	9.04
.357	9.07
.358	9.09
.366	9.30
.375	9.53
.400	10.16
.410	10.41
.411	10.44
.416	10.57
.423	10.74
.430	10.92
.451	11.46
.452	11.48
.454	11.53
.458	11.63
.474	12.04
.500	12.70
.505	12.83
.510	12.95

3. Why there are no names

Alan decided that the calibre is a diameter and nothing else (docs/NOTES-FROM-PLANNING.md entry 107 section 1), reversing entries 105 and 106, which read names through a table. Every defect those entries fixed came from reading a name as a diameter, and every name the table did not know still fell back to a guess. A diameter has one meaning. Entry 108 refused the designations above, which a person might type as a diameter with a decimal point or mm after it.